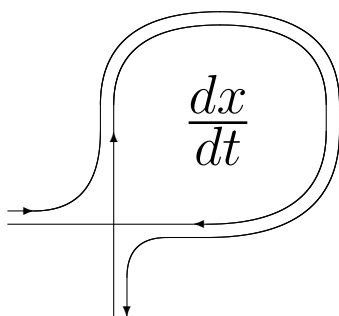




DIFFERENTIAL EQUATIONS
AND
CONTROL PROCESSES
N. 3, 2026
Electronic Journal,
reg. N Φ C77-39410 at 15.04.2010
ISSN 1817-2172
<http://diffjournal.spbu.ru/>
e-mail: jodiff@mail.ru

Control problems in nonlinear systems

Analysis and control of positive 2-D systems in continuous-time Caputo-Roesser model

N. K. Anh¹, T. T. Anh Duong¹, L. V. Hien^{1,*}, N. P. Nguyen²

¹School of Mathematics and Computer Science,
Hanoi National University of Education, Hanoi, Vietnam

²Ha Long High school for Gifted students,

*Corresponding author: hienlv@hnue.edu.vn

Abstract. In this paper, the problems of stability and stabilization via static output-feedback control are extended for positive continuous-time Roesser model described Caputo derivatives with incommensurate orders. A general solution representation to state-space equation of the model is first given. Then, using the established solution representation, necessary and sufficient conditions ensuring the positivity of the system (i.e. all system states initiated from nonnegative states are always nonnegative for all times) are derived. By utilizing the monotonicity induced from the positivity, a necessary and sufficient stability condition is formulated in terms of a tractable LP condition. The obtained stability result is then taken into deriving LP-based conditions for the existence of a desired controller that makes the closed-loop system positive and stable.

Keywords: Fractional-order 2-D systems, positivity, Roesser model, linear programming.

1 Introduction

Two-dimensional (2-D) systems are widely used to describe dynamics of various practical and physical processes where the information propagation occurs in each of the two independent directions [1]. For example, a number of partial differential equations (PDEs) with two independent variables appearing in gas absorption, heat exchangers or image processing can be converted into 2-D systems which are important for realizing control processes [2–4]. Among various models of 2-D systems, Roesser model [5] has found practical applications in numerous areas such as 2-D control systems [6], digital filter structures [7], linear repetitive processes [8, 9], or iterative learning control [10–12]. Due to a wide range of applications, the theory of 2-D systems has received significant research attention in the past few years. In particular, various problems in the systems and control theory including stability, performance analysis and controller/filter synthesis have been extensively studied for 2-D systems described by Roesser-type models. To mention a few, we refer the reader to existing works [13–19].

Recently, the problems of stability analysis and control of 2-D systems with or without delays have received remarkable attention and a number of results have been reported in the literature. While the theory of 2-D discrete-time systems has been well studied and developed, there have been only a few results devoted to the study of 2-D systems in the continuous-time domain. For example, the problems of H_∞ stabilization and reliable H_∞ stabilization via state-feedback controllers were studied for some classes of 2-D continuous-time Roesser systems with time-varying delays in [20, 21] and [22], respectively. Based on a variant of the Lyapunov–Krasovskii functional (LKF) method, sufficient conditions ensuring asymptotic stability with H_∞ performance of the closed-loop systems were derived in a linear matrix inequality (LMI) setting. The problem of delay-dependent exponential stability of 2-D continuous-time Roesser systems was also investigated in [23]. More recently, the problem of exponential stabilization of 2-D continuous-time Roesser systems with directional input delays was considered in [24] for the first time. Based on a modified reduction method and by utilizing the extended LKF method proposed in [23], analysis and synthesis conditions of a prediction-based state-feedback controller were derived in the form of tractable LMIs, which can be effectively solved by various computational tools to obtain desired controller gains.

On the other hand, fractional calculus with a long history as a pure math-

emational field is most suitable to use in description of hereditary properties of various materials and processes which are neglected in classical integer models. In particular, fractional-order (FrO) differential equations have found applications in many areas such as viscoelastic, dielectric polarization, electromagnetic waves, heat conduction, robotics or circuit systems [25–27]. In the past two decades, the study of FrO dynamical 1-D systems has attracted considerable attention with various interesting results have been reported [28–30]. Such studies for FrO 2-D systems have received considerably less research attention, especially for 2-D continuous-time systems. The notion of FrO 2-D discrete systems was introduced and studied in [31]. Continuous 2-D FrO systems in the Roesser structure and their extensions were introduced in [26, 32] and state-space equations involving partial FrO Caputo derivative were presented. Recently, the problem of positivity analysis was developed for continuous 2-D Fornasini-Marchesini fractional model in [33]. A representation of general solutions is obtained by using certain type of Laplace transformations. By employing the obtained solution formula, positivity of the system is discussed. However, due to substantial differences in their structures, the results of [33] are generally not applicable for FrO 2-D continuous Roesser systems. Extending such studies for 2-D Roesser systems proves to be more challenging and evolving, which motivates further investigation.

In this paper, we consider a class of 2-D systems described by a continuous-time Roesser model via Caputo derivatives with incommensurate orders. The addressed problem consists of the following aspects. First, we employ a 2-D Laplace transformation to formulate a general solution representation. We then derive conditions that characterize the positivity of the system. Third, by utilizing the monotonicity induced from the positivity, we exploit necessary and sufficient conditions in terms of tractable linear programming (LP) settings to ensure the system stability. Finally, the obtained stability result is utilized to address the design problem of static output-feedback controllers (SOFs) that make the corresponding closed-loop system positive and stable.

2 Preliminaries

In this section, we briefly introduce some concepts of Caputo fractional integrals and derivatives [25]. Let $L^1(0, T; X)$ be the space of integrable functions on $[0, T]$, in the Bochner sense.

Definition 1. *For a given scalar $\alpha > 0$, the fractional integral of order α of a*

function $f \in L^1(0, T; X)$ is defined by

$$I_t^\alpha f(t) = \frac{1}{\Gamma(\alpha)} \int_0^t (t-s)^{\alpha-1} f(s) ds \quad (1)$$

provided that the concerned integral is well defined, where $\Gamma(\cdot)$ is the Euler-Gamma function given by $\Gamma(p) = \int_0^\infty e^{-t} t^{p-1} dt$.

Definition 2. Let N be a positive integer. For a function $f \in C^N([0, T]; X)$, the Caputo fractional derivative of order $\alpha \in (N-1, N)$ of f is defined by

$$D_t^\alpha f(t) = \frac{1}{\Gamma(N-\alpha)} \int_0^t \frac{f^{(N)}(s)}{(t-s)^{\alpha-N+1}} ds. \quad (2)$$

Based on the concept of Caputo fractional derivative, the generalized Caputo fractional derivative of order $\alpha \in (0, 1)$ of a function f is defined as

$$D_t^{\alpha+} f(t) = \frac{1}{\Gamma(1-\alpha)} D^+ \left(\int_0^t \frac{f(s) - f(0)}{(t-s)^\alpha} ds \right),$$

where D^+ is the upper-right Dini derivative. By a simple calculus, it can be verified that

$$D_t^{\alpha+} f(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t \frac{D^+ f(s)}{(t-s)^\alpha} ds.$$

For a function $f(\cdot) \in C^1[0, \infty)$ and a scalar $0 < \alpha < 1$, the relation between the RL derivative ${}^{RL}D^\alpha f(t)$ and the Caputo derivative $D_0^\alpha f(t)$ is given by [25]

$$D_t^\alpha f(t) = {}^{RL}D^\alpha f(t) - \frac{f(0)}{\Gamma(1-\alpha)} t^{-\alpha}.$$

Definition 3. For a function f of two variables (t_1, t_2) and positive scalar α , the Caputo fractional partial derivative of order α of f with respect to t_1 is defined by

$$\partial_{t_1}^\alpha f(t_1, t_2) = \frac{1}{\Gamma(N-\alpha)} \int_0^{t_1} \frac{\frac{\partial^N}{\partial t_1^N} f(s, t_2)}{(t_1-s)^{\alpha-N+1}} ds, \quad (3)$$

where $N \in \mathbb{Z}_+$ satisfying $N-1 < \alpha < N$. The Caputo fractional partial derivative $\partial_{t_2}^\alpha f(t_1, t_2)$ of function f with respect to t_2 is defined similarly.

Let $f(t_1, t_2)$ be a continuous function of two variables (t_1, t_2) . We define 1-D Laplace transforms $F(p, t_2)$ and $F(t_1, s)$ of f with respect to t_1 and t_2 as

$$F(p, t_2) = \mathcal{L}_{t_1}[f(t_1, t_2)] = \int_0^\infty f(t_1, t_2) e^{-pt_1} dt_1,$$

$$F(t_1, s) = \mathcal{L}_{t_2}[f(t_1, t_2)] = \int_0^\infty f(t_1, t_2) e^{-st_2} dt_2,$$

and 2-D Laplace transform of f is given by

$$\begin{aligned} F(p, s) &= \mathcal{L}_{t_1, t_2}[f(t_1, t_2)] = \mathcal{L}_{t_1} \{ \mathcal{L}_{t_2}[f(t_1, t_2)] \} = \mathcal{L}_{t_2} \{ \mathcal{L}_{t_1}[f(t_1, t_2)] \} \\ &= \int_0^\infty \int_0^\infty f(t_1, t_2) e^{-(pt_1 + st_2)} dt_1 dt_2. \end{aligned} \quad (4)$$

Since

$$\mathcal{L}_{t_1}[t_1^\alpha] = \frac{\Gamma(\alpha + 1)}{p^{\alpha+1}}$$

and

$$\mathcal{L}_{t_1} \left[\frac{\partial^{N_1}}{\partial t_1^{N_1}} f(t_1, t_2) \right] = p^{N_1} F(p, t_2) - \sum_{k=1}^{N_1} p^{N_1-k} \frac{\partial^{(k-1)}}{\partial t_1^{(k-1)}} f(t_1, t_2) \Big|_{t_1=0},$$

it can be shown that the 2-D Laplace transform of partial fractional order derivative of $f(t_1, t_2)$ with respect to t_1 is given by the following formula [26]

$$\mathcal{L}_{t_1, t_2}[\partial_{t_1}^\alpha f(t_1, t_2)] = p^\alpha F(p, s) - \sum_{k=1}^{N_1} p^{\alpha-k} F_{t_1}^{(k-1)}(0, s), \quad (5)$$

where

$$F_{t_1}^{(k)}(0, s) = \mathcal{L}_{t_1} \left\{ \left[\frac{\partial^k}{\partial t_1^k} f(t_1, t_2) \right]_{t_1=0} \right\}.$$

Similarly, the 2-D Laplace transform of partial fractional order derivative of $f(t_1, t_2)$ with respect to t_2 is given by the following formula

$$\mathcal{L}_{t_1, t_2}[\partial_{t_2}^\beta f(t_1, t_2)] = s^\beta F(p, s) - \sum_{l=1}^{N_2} s^{\beta-l} F_{t_2}^{(l-1)}(p, 0), \quad (6)$$

where

$$F_{t_2}^{(l)}(p, 0) = \mathcal{L}_{t_2} \left\{ \left[\frac{\partial^l}{\partial t_2^l} f(t_1, t_2) \right]_{t_2=0} \right\}.$$

3 Solution representation

Consider the following FrO 2-D continuous-time Roesser system

$$\begin{bmatrix} \partial_{t_1}^\alpha x^h(t_1, t_2) \\ \partial_{t_2}^\beta x^v(t_1, t_2) \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u(t_1, t_2), \quad (7a)$$

$$y(t_1, t_2) = C \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} + Du(t_1, t_2), \quad (7b)$$

where $x^h \in \mathbb{R}^{n_h}$ and $x^v \in \mathbb{R}^{n_v}$ are the horizontal and vertical state vectors, respectively, A_{ij} ($i, j = 1, 2$), B_k ($k = 1, 2$) and C, D are given real matrices of appropriate dimensions, $u(t_1, t_2) \in \mathbb{R}^m$ and $y(t_1, t_2) \in \mathbb{R}^q$ are the control input and measurement output, respectively; $0 < \alpha < 1$ and $0 < \beta < 1$ are scalars representing the order of partial derivatives. Initial conditions of (7) are specified as

$$x^h(0, t_2) = \phi(t_2), \quad x^v(t_1, 0) = \psi(t_1), \quad t_1, t_2 \in \mathbb{R}_+. \quad (8)$$

By applying 2-D Laplace transforms (5) and (6) to (7a), we obtain

$$\begin{bmatrix} p^\alpha X^h(p, s) - p^{\alpha-1} X^h(0, s) \\ s^\beta X^v(p, s) - s^{\beta-1} X^v(p, 0) \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} X^h(p, s) \\ X^v(p, s) \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} U(p, s), \quad (9)$$

where $X^h(p, s)$, $X^v(p, s)$ and $U(p, s)$ are the 2-D Laplace transforms of $x^h(t_1, t_2)$, $x^v(t_1, t_2)$ and $u(t_1, t_2)$, respectively, $X^h(0, s) = \mathcal{L}_{t_2}[\phi(t_2)]$, $X^v(p, 0) = \mathcal{L}_{t_1}[\psi(t_1)]$. By pre-multiplying (9) with $\text{diag}\{p^{-\alpha} I_{n_h}, s^{-\beta} I_{n_v}\}$, we then obtain

$$\begin{bmatrix} X^h(p, s) \\ X^v(p, s) \end{bmatrix} = H^{-1}(p, s) \left\{ \begin{bmatrix} p^{-1} X^h(0, s) \\ s^{-1} X^v(p, 0) \end{bmatrix} + \begin{bmatrix} p^{-\alpha} B_1 \\ s^{-\beta} B_2 \end{bmatrix} U(p, s) \right\}, \quad (10)$$

where

$$H(p, s) = \begin{bmatrix} I_{n_h} - p^{-\alpha} A_{11} & -p^{-\alpha} A_{12} \\ -s^{-\beta} A_{21} & I_{n_v} - s^{-\beta} A_{22} \end{bmatrix}.$$

We now decompose

$$H^{-1}(p, s) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} T_{ij} p^{-i\alpha} s^{-j\beta} \quad (11)$$

where T_{ij} are matrix coefficients which will be determined. By the expression of the matrix $H(p, s)$ and the decomposition (11), it follows from the fact

$$H^{-1}(p, s)H(p, s) = H(p, s)H^{-1}(p, s) = I_n$$

that

$$\sum_{i=0}^{\infty} \sum_{j=0}^{\infty} \{T_{ij} - T_{10}T_{i-1,j} - T_{01}T_{i,j-1}\} p^{-i\alpha} s^{-j\beta} = I_n, \quad (12)$$

where

$$T_{10} = \begin{bmatrix} A_{11} & A_{12} \\ 0 & 0 \end{bmatrix}, \quad T_{01} = \begin{bmatrix} 0 & 0 \\ A_{21} & A_{22} \end{bmatrix}.$$

By matching the coefficients at the same powers of p and s , we obtain the following iteration

$$T_{ij} = \begin{cases} I_n & \text{for } i = j = 0, \\ T_{10}T_{i-1,j} + T_{01}T_{i,j-1} & \text{for } i + j > 0, \\ 0 & \text{for } i < 0 \text{ or } j < 0, \end{cases} \quad (13)$$

Combining (10) and (11) we obtain

$$\begin{bmatrix} X^h(p, s) \\ X^v(p, s) \end{bmatrix} = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} T_{ij} \left\{ \begin{bmatrix} p^{-i\alpha-1} s^{-j\beta} X^h(0, s) \\ p^{-i\alpha} s^{-j\beta-1} X^v(p, 0) \end{bmatrix} + \begin{bmatrix} p^{-(i+1)\alpha} s^{-j\beta} B_1 \\ p^{-i\alpha} s^{-(j+1)\beta} B_2 \end{bmatrix} U(p, s) \right\}, \quad (14)$$

where T_{ij} are uniquely determined by (13).

We now apply the 2-D inverse Laplace transformation to (14). Observe that, for $\alpha > 0$, $\beta > 0$, we have

$$\begin{aligned} \mathcal{L}^{-1}[p^{-\alpha} F(p)] &= I_t^\alpha f(t), \quad \mathcal{L}_{t_1}^{-1}[p^{-\alpha}] = \frac{t_1^{\alpha-1}}{\Gamma(\alpha)}, \\ \mathcal{L}_{t_1, t_2}^{-1}[p^{-\alpha} F(p, s)] &= I_{t_1}^\alpha f(t_1, t_2) = \frac{1}{\Gamma(\alpha)} \int_0^{t_1} \frac{f(\theta, t_2)}{(t_1 - \theta)^{1-\alpha}} d\theta, \\ \mathcal{L}_{t_1, t_2}^{-1}[s^{-\beta} F(p, s)] &= I_{t_2}^\beta f(t_1, t_2) = \frac{1}{\Gamma(\beta)} \int_0^{t_2} \frac{f(t_1, \theta)}{(t_2 - \theta)^{1-\beta}} d\theta, \end{aligned} \quad (15)$$

and

$$\mathcal{L}_{t_1, t_2}^{-1}[p^{-\alpha} s^{-\beta} F(p, s)] = \frac{1}{\Gamma(\alpha)\Gamma(\beta)} \int_0^{t_1} \int_0^{t_2} \frac{f(\xi, \theta)}{(t_1 - \xi)^{1-\alpha}(t_2 - \theta)^{1-\beta}} d\xi d\theta. \quad (16)$$

By utilizing the properties (15), (16) of 2-D inverse Laplace transformation, from (14), we obtain the following solution representation of the problem gov-

erned by (7a) and (8)

$$\begin{aligned} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} = & \sum_{i=0}^{\infty} \sum_{j=1}^{\infty} T_{ij} \left\{ \frac{t_1^{i\alpha}}{\Gamma(i\alpha + 1)} I_{t_2}^{j\beta} \begin{bmatrix} \phi(t_2) \\ 0 \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} I_{t_1, t_2}^{(i+1)\alpha, j\beta} u(t_1, t_2) \right\} \\ & + \sum_{i=0}^{\infty} T_{i0} \left\{ \frac{t_1^{i\alpha}}{\Gamma(i\alpha + 1)} \begin{bmatrix} \phi(t_2) \\ 0 \end{bmatrix} + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} I_{t_1}^{(i+1)\alpha} u(t_1, t_2) \right\} \\ & + \sum_{i=1}^{\infty} \sum_{j=0}^{\infty} T_{ij} \left\{ \frac{t_2^{j\beta}}{\Gamma(j\beta + 1)} I_{t_1}^{i\alpha} \begin{bmatrix} 0 \\ \psi(t_1) \end{bmatrix} + \begin{bmatrix} 0 \\ B_2 \end{bmatrix} I_{t_1, t_2}^{i\alpha, (j+1)\beta} u(t_1, t_2) \right\} \\ & + \sum_{j=0}^{\infty} T_{0j} \left\{ \frac{t_2^{j\beta}}{\Gamma(j\beta + 1)} \begin{bmatrix} 0 \\ \psi(t_1) \end{bmatrix} + \begin{bmatrix} 0 \\ B_2 \end{bmatrix} I_{t_2}^{(j+1)\beta} u(t_1, t_2) \right\}. \end{aligned} \quad (17)$$

We summarize the above derivation in the following theorem.

Theorem 1. *For given continuous functions ϕ , ψ , real numbers $0 < \alpha < 1$, $0 < \beta < 1$, and for an arbitrary input $u(t_1, t_2)$, the solution of the problem governed by system (7a) and (8) is given by (17), where the matrix coefficients T_{ij} , $i, j \geq 0$, are determined by (13).*

4 Positivity

In this section, we will derive conditions by which a fractional order 2-D system in the form of (7) is positive. First, we give the following definition.

Definition 4. *System (7) is said to be internally positive if state and output trajectories are always nonnegative for any nonnegative initial conditions and control input. In other words, if $\phi(t_2) \succeq 0$, $\psi(t_1) \succeq 0$ and $u(t_1, t_2) \succeq 0$ for all $t_1 \geq 0, t_2 \geq 0$, then it holds that $x(t_1, t_2) = \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} \succeq 0$ and $y(t_1, t_2) \succeq 0$ for all $(t_1, t_2) \in \mathbb{R}_+^2$.*

Before proceeding further, we reformulate the following auxiliary results.

Proposition 1. *The 1-D fractional-order system*

$$\begin{aligned} D_t^\alpha x(t) &= Ax(t) + Bu(t), \quad t > 0, \\ x(0) &= x_0 \in \mathbb{R}^n, \end{aligned} \quad (18)$$

is internally positive if and only if A is Metzler and B is nonnegative.

Proof. We now present a brief proof for the sufficiency. Since A is a Metzler matrix, there exists a scalar $\sigma > 0$ such that $A^\sigma \triangleq A + \sigma I_n \succeq 0$. Then, we have

$$\begin{aligned} x(t) = E_\alpha(-\sigma t^\alpha)x_0 + \int_0^t \Psi(t-s)A^\sigma x(s)ds \\ + \int_0^t \Psi(t-s)Bu(s)ds, \end{aligned} \quad (19)$$

where $\Psi(t) = \sum_{k=0}^{\infty} \frac{-\sigma^k t^{(k+1)\alpha-1}}{\Gamma(k\alpha+\alpha)} = t^{\alpha-1}E_{\alpha,\alpha}(-\sigma t^\alpha)$. Note that $E_\alpha(-\sigma t^\alpha) > 0$ and $\Psi(t) > 0$ for all $t > 0$ and $\alpha \in (0, 1)$ by the complete monotonicity of the functions $E_\alpha(-z)$ and $E_{\alpha,\alpha}(-z)$ on $[0, \infty)$. Thus, for any $x_0 \in \mathbb{R}_+^n$ and nonnegative input $u(t) \in \mathbb{R}_+^m$, $x(t) \in \mathbb{R}_+^n$ for small $t > 0$. Suppose that there exists a $t_* > 0$ such that $x(t_*) \not\succeq 0$ and $x(t) \succeq 0$ for all $t \in [0, t_*)$. Then, it can be deduced from (19) that $x(t) \succeq E_\alpha(-\sigma t^\alpha)x_0$, $t \in [0, t_*)$, which yields $x(t_*) \succeq 0$ by letting $t \uparrow t_*$. This contradiction shows that $x(t) \succeq 0$ for all $t \geq 0$. The proof is completed.

Lemma 1. For a given $0 < \alpha < 1$, the matrix $E_\alpha(At^\alpha) \succeq 0$ for all $t > 0$ if and only if A is Metzler, where $E_\alpha(z)$ denotes the one-parameter Mittag-Leffler function,

$$E_\alpha(z) = \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(\alpha k + 1)}.$$

Proof. If A is Metzler then, for zero input (i.e. $u(t) = 0$) and $x(0) = e_k$, where e_k is the k th column of the identity matrix I_n , we have $x_k(t) = E_\alpha(At^\alpha)e_k \succeq 0$ according to Proposition 1. Thus,

$$E_\alpha(At^\alpha) = [x_1(t) \ x_2(t) \cdots x_n(t)] \succeq 0,$$

which proves the sufficiency.

For the necessity, assume that $E_\alpha(At^\alpha) \succeq 0$ for all $t > 0$. If there exists an entry $a_{ij} < 0$, $i \neq j$, then we have

$$0 \leq e_i^\top E_\alpha(At^\alpha)e_j = \frac{a_{ij}t^\alpha}{\Gamma(\alpha + 1)} + o(t^\alpha), \quad t \rightarrow 0.$$

Since $\lim_{t \downarrow 0} o(t^\alpha)/t^\alpha = 0$, for $\epsilon = -a_{ij}/(2\Gamma(\alpha + 1)) > 0$, there exists a $\delta > 0$ such that $|o(t^\alpha)|/t^\alpha < \epsilon$, $t \in (0, \delta)$. Therefore,

$$e_i^\top E_\alpha(At^\alpha)e_j < \frac{a_{ij}t^\alpha}{2\Gamma(\alpha + 1)} < 0, \quad t \in (0, \delta).$$

This contradiction implies that A is a Metzler matrix. The proof is completed.

Theorem 2. The 2-D system (7) is internally positive if and only if $\mathcal{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ is Metzler and $\mathcal{B} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, C , D are nonnegative.

Proof. Necessity: For zero input (i.e. $u = 0$) and $\phi = 0$, it follows from solution formula (17) that

$$\begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} = \sum_{i=1}^{\infty} \sum_{j=0}^{\infty} T_{ij} \frac{t_2^{j\beta}}{\Gamma(j\beta + 1)} I_{t_1}^{i\alpha} \begin{bmatrix} 0 \\ \psi(t_1) \end{bmatrix} + \sum_{j=0}^{\infty} T_{0j} \frac{t_2^{j\beta}}{\Gamma(j\beta + 1)} \begin{bmatrix} 0 \\ \psi(t_1) \end{bmatrix}. \quad (20)$$

For sufficiently small $t_1 > 0$, by the continuity of fractional integrals, from (20), we have

$$\begin{aligned} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} &\approx \sum_{j=0}^{\infty} T_{0j} \frac{t_2^{j\beta}}{\Gamma(j\beta + 1)} \begin{bmatrix} 0 \\ \psi(t_1) \end{bmatrix} \\ &= \sum_{j=0}^{\infty} \frac{t_2^{j\beta} T_{01}^j}{\Gamma(j\beta + 1)} \begin{bmatrix} 0 \\ \psi(t_1) \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ \sum_{j=0}^{\infty} \frac{t_2^{j\beta} A_{22}^j}{\Gamma(j\beta + 1)} \psi(t_1) \end{bmatrix} = \begin{bmatrix} 0 \\ E_{\beta}(A_{22} t_2^{\beta}) \psi(t_1) \end{bmatrix}. \end{aligned} \quad (21)$$

Let $\psi(t_1)$ be a continuous function such that $\psi(t_1) \succeq 0$ for all $t_1 \geq 0$ and $\psi(t_1) \succ 0$ for sufficiently small $t_1 > 0$. Then, it follows from $\begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} \succeq 0$ for all $t_1, t_2 \geq 0$ that $E_{\beta}(A_{22} t_2^{\beta}) \succeq 0$ for all $t_2 \geq 0$. By Lemma 1, we can conclude that the matrix A_{22} is Metzler. It can be shown by similar arguments that, for $u = 0, \psi = 0$, and $\phi(t_2) \succ 0$ for $t_2 > 0$, A_{11} is also a Metzler matrix.

On the other hand, for zero input $u = 0$, similar to (21), we have

$$\begin{aligned} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} &= \begin{bmatrix} E_{\alpha}(t_1^{\alpha} A_{11}) \phi(t_2) \\ E_{\beta}(t_2^{\beta} A_{22}) \psi(t_1) \end{bmatrix} \\ &+ \sum_{i=0}^{\infty} \sum_{j=1}^{\infty} \frac{t_1^{i\alpha}}{\Gamma(1 + i\alpha)} (T_{10} T_{i-1,j} + T_{01} T_{i,j-1}) \begin{bmatrix} I_{t_2}^{j\beta} \phi(t_2) \\ 0 \end{bmatrix} \\ &+ \sum_{i=1}^{\infty} \sum_{j=0}^{\infty} \frac{t_2^{j\beta}}{\Gamma(1 + j\beta)} (T_{10} T_{i-1,j} + T_{01} T_{i,j-1}) \begin{bmatrix} 0 \\ I_{t_1}^{i\alpha} \psi(t_1) \end{bmatrix}. \end{aligned} \quad (22)$$

For sufficiency small $t_1 > 0$, $t_2 > 0$, and keep in mind that the matrices A_{11}, A_{22} are Metzler, we obtain the necessity that the matrices A_{12} and A_{21} are nonnegative.

Finally, for zero initial condition and $u(t_1, t_2) \succ 0$, it follows from (17) that

$$x(t_1, t_2) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} T_{i-1, j-1} B I_{t_1, t_2}^{i\alpha, j\beta} u(t_1, t_2). \quad (23)$$

For sufficiently small t_1, t_2 , we have

$$x(t_1, t_2) \approx B I_{t_1, t_2}^{\alpha, \beta} u(t_1, t_2).$$

In combination with the fact the fractional integral $I_{t_1, t_2}^{\alpha, \beta} u(t_1, t_2) \succeq 0$ for $u(t_1, t_2) \succeq 0$, we can conclude that $B \succeq 0$. Finally, the condition that $C, D \succeq 0$ is shown by specifying special initial conditions and inputs.

5 Stability analysis

In this section, we focus on the stability of positive 2-D systems in the form of (7). Assume that the matrix $\mathcal{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ is Metzler and $B = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$, C , D are nonnegative. By Theorem 2, system (7) is positive. With $u = 0$, the open-loop system of (7) is given as

$$\begin{bmatrix} \partial_{t_1}^{\alpha} x^h(t_1, t_2) \\ \partial_{t_2}^{\beta} x^v(t_1, t_2) \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix}. \quad (24)$$

Let $x(t_1, t_2)$ and $\tilde{x}(t_1, t_2)$ be solutions of (24) with initial conditions (ϕ, ψ) and $(\tilde{\phi}, \tilde{\psi})$, respectively. Since $\hat{x}(t_1, t_2) = \tilde{x}(t_1, t_2) - x(t_1, t_2)$ is a solution of (24), if $\psi(t_1) \preceq \tilde{\psi}(t_1)$ and $\phi(t_2) \preceq \tilde{\phi}(t_2)$ for all $t_1 \geq 0, t_2 \geq 0$, then $\hat{x}(t_1, t_2) \succeq 0$ for all $(t_1, t_2) \in \mathbb{R}_+^2$ according to the positivity of (24). This shows that

$$|x(t_1, t_2)| \preceq x(t_1, t_2, |\phi|, |\psi|), \quad t_1 \geq 0, \quad t_2 \geq 0,$$

where $x(t_1, t_2, |\phi|, |\psi|)$ is the solution of (24) with initial condition $|\phi|, |\psi|$. It is clear that $|\psi(t_1)| \succeq 0$ and $|\phi(t_2)| \succeq 0$ for all $t_1 \geq 0, t_2 \geq 0$. In addition, according to the above estimate, it can be verified that $x(t_1, t_2) \rightarrow 0$ as $t_1 \rightarrow \infty$ and $t_2 \rightarrow \infty$ for all initial functions ϕ, ψ if and only if it holds for all nonnegative initial functions $\psi(t_1) \succeq 0$ and $\phi(t_2) \succeq 0$. Therefore, in the following, it suffices

to consider nonnegative initial functions ϕ, ψ . In regard to the positivity of system (24), we have $x(t_1, t_2) \succeq 0$ for all $t_1 \geq 0, t_2 \geq 0$.

We recall here that system (24) is said to be globally asymptotically stable (GAS) with respect to an admissible set of initial conditions if for any initial conditions ϕ, ψ , it holds that $x(t_1, t_2) \rightarrow 0$ as $t_1 \rightarrow \infty$ and $t_2 \rightarrow \infty$. In this paper, an admissible set of initial conditions is defined by

$$\mathcal{A}_{\mathcal{F}} = \left\{ (\phi, \psi) \in C(\mathbb{R}_+, \mathbb{R}^{n_h} \times \mathbb{R}^{n_v}) : \sup_{t \geq 0} \left\| \begin{bmatrix} \phi(t) \\ \psi(t) \end{bmatrix} \right\| < \infty \right\}.$$

We now derive necessary stability conditions. Assume that positive system (24) is GAS and there exists a vector $\xi = \begin{bmatrix} \xi_h \\ \xi_v \end{bmatrix} \in \mathbb{R}_+^{n_h} \times \mathbb{R}_+^{n_v}, \xi \neq 0$, such that $\mathcal{A}\xi \succeq 0$. Let $x_\xi(t_1, t_2)$ be the solution of (24) with constant initial conditions $\phi(t_2) = \xi_h, \psi(t_1) = \xi_v$, and define the function $x(t_1, t_2) = x_\xi(t_1, t_2) - \xi$. It follows from (24) that

$$\begin{aligned} \begin{bmatrix} \partial_{t_1}^\alpha x^h(t_1, t_2) \\ \partial_{t_2}^\beta x^v(t_1, t_2) \end{bmatrix} &= \underbrace{\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} x_\xi^h(t_1, t_2) \\ x_\xi^v(t_1, t_2) \end{bmatrix} \\ &= \mathcal{A}[x_\xi(t_1, t_2) - \xi] + \mathcal{A}\xi. \end{aligned} \quad (25)$$

Since $\mathcal{A}\xi \succeq 0$ and, according to (24), $x(t_1, t_2)$ is the solution of (7) with zero initial condition, it holds that $x(t_1, t_2) \succeq 0$ for all $(t_1, t_2) \in \mathbb{R}_+^2$. Therefore,

$$0 \neq \xi \preceq x_\xi(t_1, t_2) \rightarrow 0, \quad t_1, t_2 \rightarrow \infty.$$

This contradiction shows that for any nonzero vector $\xi \in \mathbb{R}_+^{n_h+n_v}$, there exists at least a component of the vector $\mathcal{A}\xi$ that is strictly negative (i.e. $(\mathcal{A}\xi)_k < 0$ for some $k = 1, 2, \dots, n$). By similar approach utilizing the KKM lemma (see [35] for more details), we obtain the following necessary stability condition.

Proposition 2. *Assume that the positive FrO 2-D system (24) is GAS. Then, there must exist a vector $\xi = \begin{bmatrix} \xi_h \\ \xi_v \end{bmatrix} \in \mathbb{R}^{n_h} \times \mathbb{R}^{n_v}, \xi \succ 0$, such that*

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \xi_h \\ \xi_v \end{bmatrix} \prec 0. \quad (26)$$

We now in a position to derive necessary and sufficient LP-based stability condition for positive fractional-order 2-D continuous systems.

Theorem 3. *The positive FrO 2-D system (24) is GAS if and only if there exists a vector $\xi \in \mathbb{R}^{n_h+n_v}$, $\xi \succ 0$, that satisfies the LP-based condition (26).*

Proof. The necessity follows from Proposition 2. We now prove the sufficiency.

Let $0 \prec \xi = \begin{bmatrix} \xi_h \\ \xi_v \end{bmatrix} \in \mathbb{R}^{n_h} \times \mathbb{R}^{n_v}$ be a vector satisfying (26) and $x(t_1, t_2)$ be a solution of positive system (24) with initial functions ϕ, ψ . Then, there exists a $\delta > 0$ such that

$$\phi(t_2) \preceq \delta \xi_h, \quad \psi(t_1) \preceq \delta \xi_v$$

for all $t_1, t_2 \geq 0$. Similar to the proof of Proposition 2, we can obtain

$$|x(t_1, t_2)| \preceq \delta x_\xi(t_1, t_2), \quad t_1 \geq 0, \quad t_2 \geq 0.$$

For a sufficiently small $\epsilon > 0$ such that $\mathcal{A}(\xi + \epsilon \mathbf{1}_n) \prec 0$, where $\mathbf{1}_n$ denotes the vector in $\mathbb{R}^n$ with all entries equal one, we define the function $\hat{x}(t_1, t_2) = \xi + \epsilon \mathbf{1}_n - x_\xi(t_1, t_2)$. It follows from (23) that

$$\begin{aligned} \begin{bmatrix} \partial_{t_1}^\alpha \hat{x}^h(t_1, t_2) \\ \partial_{t_2}^\beta \hat{x}^v(t_1, t_2) \end{bmatrix} &= -\mathcal{A} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} \\ &= \mathcal{A} \hat{x}(t_1, t_2) - \mathcal{A}(\xi + \epsilon \mathbf{1}_n). \end{aligned} \quad (27)$$

System (27) is positive and

$$\begin{aligned} \hat{x}^h(0, t_2) &= \xi_h + \epsilon \mathbf{1}_{n_h} - x_\xi^h(0, t_2) = \epsilon \mathbf{1}_{n_h} \succ 0, \\ \hat{x}^v(t_1, 0) &= \xi_v + \epsilon \mathbf{1}_{n_v} - x_\xi^v(t_1, 0) = \epsilon \mathbf{1}_{n_v} \succ 0, \end{aligned}$$

thus, $\hat{x}(t_1, t_2) \succeq 0$, which yields $x_\xi(t_1, t_2) \preceq \xi + \epsilon \mathbf{1}_n$ for all $t_1, t_2 \geq 0$. This shows that there exists a finite value $x_\infty = \limsup_{t_1, t_2 \rightarrow \infty} x_\xi(t_1, t_2)$ and, therefore,

$$\limsup_{t_1, t_2 \rightarrow \infty} D_{t_1, t_2}^{\alpha, \beta} x_\xi(t_1, t_2) = \mathcal{A} x_\infty,$$

where $D_{t_1, t_2}^{\alpha, \beta} x_\xi(t_1, t_2) = \begin{bmatrix} \partial_{t_1}^\alpha x_\xi^h(t_1, t_2) \\ \partial_{t_2}^\beta x_\xi^v(t_1, t_2) \end{bmatrix}$. In addition, by the properties of Laplace transformations, we have

$$\lim_{p, s \rightarrow 0} \mathbb{I}(p, s) X_\xi(p, s) = x_\infty, \quad \lim_{p, s \rightarrow 0} \mathbb{I}(p, s) \mathcal{L}_{t_1, t_2} [D_{t_1, t_2}^{\alpha, \beta} x_\xi(t_1, t_2)] = \mathcal{A} x_\infty,$$

where $\mathbb{I}(p, s) = \text{diag}\{pI_{n_h}, sI_{n_v}\}$. Note that

$$\mathcal{L}_{t_1, t_2}[D_{t_1, t_2}^{\alpha, \beta} x_\xi(t_1, t_2)] = \begin{bmatrix} p^\alpha X_\xi^h(p, s) - p^{\alpha-1} \xi_h \\ s^\beta X^v(p, s) - s^{\beta-1} \xi_v \end{bmatrix}.$$

Finally, we get

$$\mathcal{A}x_\infty = \lim_{p, s \rightarrow 0} \begin{bmatrix} p^\alpha (pX_\xi^h(p, s) - \xi_h) \\ s^\beta (sX^v(p, s) - \xi_v) \end{bmatrix} = 0,$$

which yields $x_\infty = 0$ and, therefore, system (24) is GAS. The proof is completed.

6 Controller synthesis

Consider the following control system described by a 2-D FrO Roesser model

$$\begin{bmatrix} \partial_{t_1}^\alpha x^h(t_1, t_2) \\ \partial_{t_2}^\beta x^v(t_1, t_2) \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix} + \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} u(t_1, t_2), \quad (28a)$$

$$y(t_1, t_2) = C \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix}, \quad (28b)$$

where $u(t_1, t_2)$ is the control input and $y(t_1, t_2)$ is the measurement output. By

Theorem 2, system (28) is positive if and only if the matrix $\mathcal{A} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$ is

Metzler and $\mathcal{B} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} \succeq 0$, $C \succeq 0$. In the following, we assume that system (28) is positive. An SOFC will be designed in the form

$$u(t_1, t_2) = Ky(t_1, t_2) = KCx(t_1, t_2), \quad (29)$$

where $K \in \mathbb{R}^{m \times q}$ is the controller gain which will be determined. By incorporating (29), the closed-loop system of (28) is obtained as

$$\begin{bmatrix} \partial_{t_1}^\alpha x^h(t_1, t_2) \\ \partial_{t_2}^\beta x^v(t_1, t_2) \end{bmatrix} = \underbrace{(\mathcal{A} + \mathcal{B}KC)}_{\mathcal{A}_c} \begin{bmatrix} x^h(t_1, t_2) \\ x^v(t_1, t_2) \end{bmatrix}. \quad (30)$$

By Theorems 2 and 3, the closed-loop system (30) is positive and GAS if and only if the following conditions hold

$$\mathcal{A}_c = \mathcal{A} + \mathcal{B}KC \text{ is Metzler,} \quad (31a)$$

$$\exists \eta \succ 0 : \eta^\top (\mathcal{A} + \mathcal{B}KC) \prec 0. \quad (31b)$$

Let $\mathcal{B} = [b_1 \ b_2 \ \cdots \ b_m]$, $b_j \in \mathbb{R}^n$, and $K^\top = [k_1 \ k_2 \ \cdots \ k_m]$, $k_j \in \mathbb{R}^q$. Then, for any $\eta \in \mathbb{R}^n$, $\eta \succ 0$, we have

$$\eta^\top \mathcal{B}K = \sum_{j=1}^m (b_j^\top \eta) k_j^\top.$$

By using the change of variable $k_j^\top = \frac{1}{\sigma} z_j^\top$, where $z_j \in \mathbb{R}^q$, $\sigma > 0$ is a constant, and under additional condition that $z_j \succeq z$ for all j , condition (31b) is reduced to

$$\eta^\top \mathcal{A} + \frac{1}{\sigma} \sum_{j=1}^m (b_j^\top \eta) z_j^\top C \prec 0. \quad (32)$$

To get decoupling of design parameters η and z in (32), we take $\sigma = \sum_{j=1}^m (b_j^\top \eta) = \mathbf{1}_m^\top \mathcal{B}^\top \eta$. Then, $\mathcal{B}K = \frac{1}{\sigma} \sum_{j=1}^m b_j z_j^\top$. It is well-known that a matrix $M \in \mathbb{R}^{n \times n}$ is Metzler if and only if there exists a constant ς such that $M + \varsigma I_n \succeq 0$. Therefore, with $\mathcal{B}K = \frac{1}{\sigma} \mathcal{B}Z^\top$, where $\sigma = \mathbf{1}_m^\top \mathcal{B}^\top \eta$, the matrix $\mathcal{A}_c = \mathcal{A} + \mathcal{B}KC$ in (31a) can be written as $\mathcal{A}_c = \frac{1}{\sigma} ((\mathbf{1}_m^\top \mathcal{B}^\top \eta) \mathcal{A} + \mathcal{B}Z^\top C)$. By changing the variable $\hat{\sigma} = \varsigma \sigma$, condition (31a) is satisfied if and only if

$$(\mathbf{1}_m^\top \mathcal{B}^\top \eta) \mathcal{A} + \mathcal{B}Z^\top C + \hat{\sigma} I_n \succeq 0.$$

In summary, the derived conditions in (31) are guaranteed by the following LP-based conditions

$$\begin{cases} (\mathbf{1}_m^\top \mathcal{B}^\top \eta) \mathcal{A} + \mathcal{B}Z^\top C + \hat{\sigma} I_n \succeq 0 \\ \eta^\top \mathcal{A} + z^\top C \prec 0 \\ Z \succeq \mathbf{1}_m z^\top \\ \eta \in \mathbb{R}^n, \eta \succ 0, \hat{\sigma} \in \mathbb{R}_+, Z \in \mathbb{R}^{m \times q}, z \in \mathbb{R}^q. \end{cases} \quad (33)$$

7 An illustrative example

Consider the following fractional-order model of control systems which can be used in describing various dynamical processes in gas absorption, water stream

heating, or air drying [36]

$$\begin{aligned} \begin{bmatrix} \partial_x^\alpha r(x, t) \\ \partial_t^\beta s(x, t) \end{bmatrix} &= \begin{bmatrix} a_1 & a_1 a_2 + a_0 \\ 1 & a_2 \end{bmatrix} \begin{bmatrix} r(x, t) \\ s(x, t) \end{bmatrix} + \begin{bmatrix} b \\ 0 \end{bmatrix} f(x, t), \\ y(x, t) &= [0 \quad 1] \begin{bmatrix} r(x, t) \\ s(x, t) \end{bmatrix}, \end{aligned} \quad (34)$$

where a_0 , a_1 , a_2 , and b are real coefficients and $f(x, t)$ is the input function. The following parameters are borrowed from [36]

$$a_0 = 0.2, \quad a_1 = -3.0, \quad a_2 = -1.0, \quad b = 0.3.$$

Then, we have

$$\mathcal{A} = \begin{bmatrix} -3.0 & 3.2 \\ 1.0 & -1.0 \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} 0.3 \\ 0 \end{bmatrix}.$$

In system (34), the input function $u = f(x, t)$ can be regarded as control input signal. Since $\mathcal{A}$ is Metzler and $\mathcal{B} \succeq 0$, by Theorem 2, system (34) is positive. Moreover, it is easy to verify that the LP-based condition (26) does not deliver any feasible solution $\xi \succ 0$. Thus, the unforced system (i.e. $u = 0$) is unstable. Simulation results presented in Figs. 1 and 2 are taken with $\alpha = \beta = 0.5$ and initial conditions $\phi(t) = 1$, $\psi(x) = 1$ for $t, x \in [0, 50]$. The obtained simulations demonstrate that the open-loop system is unstable.

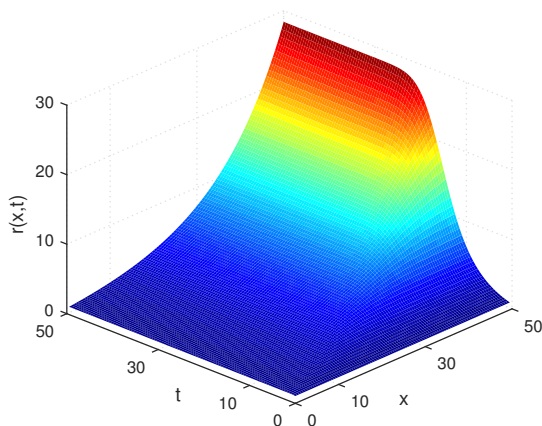


Figure 1: Open-loop state trajectory $r(x, t)$

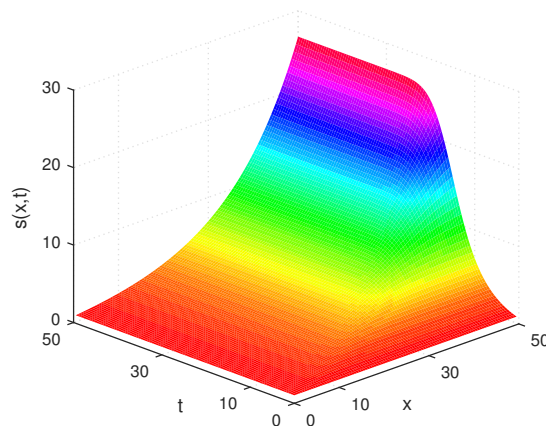


Figure 2: Open-loop state trajectory $s(x, t)$

We now apply the the design conditions proposed in this paper. By using the **LinProg** tool in Matlab, it can be verified that the LP-based conditions in (33) are feasible with $\eta = \begin{bmatrix} 0.8191 \\ 0.6284 \end{bmatrix}$ (some other parameters are omitted) and

the gain K is obtained as $K = -9.5782$. It is clear that the closed-loop system matrix

$$\mathcal{A}_c = \mathcal{A} + \mathcal{B}KC = \begin{bmatrix} -3.0 & 0.3265 \\ 1.0 & -1.0 \end{bmatrix}$$

is Metzler and Hurwitz. Thus, the closed-loop system is positive and stable. Simulation results with $\alpha = \beta = 0.5$ and initial conditions $\phi(t) = 2$, $\psi(x) = 2$ for $t, x \in [0, 50]$ are given in Figs. 3 and 4. It can be seen that the closed-loop trajectories converge to the origin as revealed by the obtained theoretical results. This illustrates the effectiveness of the proposed design conditions.

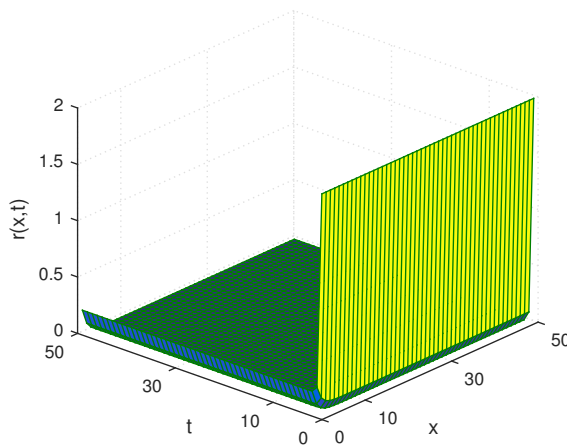


Figure 3: Closed-loop state trajectory $r(x, t)$

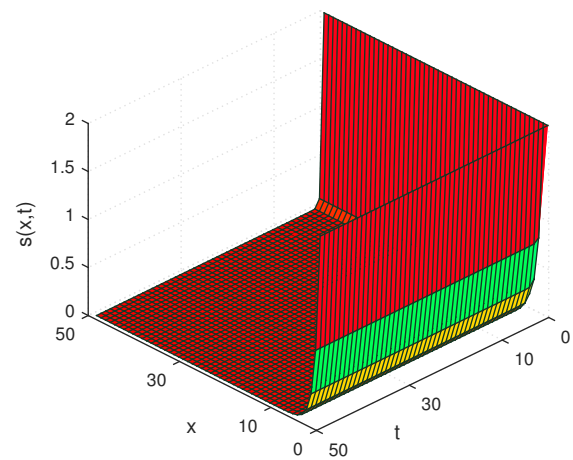


Figure 4: Closed-loop state trajectory $s(x, t)$

8 Conclusion

The problems of positivity, stability analysis and static output-feedback controller design have been extended for continuous-time Roesser model described Caputo fractional-order derivatives. A general solution representation to state-space equation of the model has been obtained using 2-D Laplace transformations and the positivity of the system has been discussed. It is shown that a fractional-order continuous-time 2-D Roesser system is positive if and only if the state matrix is Metzler and other system matrices are nonnegative. The positivity has been then utilized to derive LP-stability and stabilization conditions. Numerical simulations have been provided to validate the effectiveness of the proposed method in this paper.

References

- [1] T. Kaczorek, *Two-Dimensional Linear Systems*, Berlin, Springer-Verlag, 1985.
- [2] W. Marszałek and G.T. Kekkeris, Heat exchangers and linear image processing theory, *Int. J. Heat Mass Trans.*, **32**(1989), 2363–2374.
- [3] D. Bors and S. Walczak, Application of 2D systems to investigation of a process of gas filtration, *Multidim. Syst. Signal Process.*, **23**(2012), 119–130.
- [4] A. Alenany, G. Mercère and J.A. Ramos, Subspace identification of 2-D CRSD Roesser models with deterministic-stochastic inputs: A state computation approach, *IEEE Trans. Control Syst. Techno.*, **25**(2017), 1108–1115.
- [5] R.P. Roesser, A discrete state-space model for linear image processing, *IEEE Trans. Autom. Control*, **20**(1975), 1–10.
- [6] E. Rogers, K. Galkowski, W. Paszke, K.L. Moore, P.H. Bauer, L. Hladowski and P. Dabkowski, Multidimensional control systems: case studies in design and evaluation, *Multidim. Syst. Signal Process.*, **26**(2015), 895–939.
- [7] L. Wu and Z. Wang, *Filtering and Control for Classes of Two-Dimensional Systems*, Switzerland, Springer, 2015.
- [8] L. Zhou, J. She, S. Zhou and C. Li, Compensation for state-dependent nonlinearity in a modified repetitive control system, *Int. J. Robust Nonlinear Control*, **28**(2018), 213–226.
- [9] L. Zhou, J. She, S. Zhou, Robust H_∞ control of an observer-based repetitive-control system, *J. Frankl. Inst.*, **355**(2018), 4952–4969.
- [10] J.E. Kurek and M.B. Zaremba, Iterative learning control synthesis based on 2-D system theory, *IEEE Trans. Autom. Control*, **38**(1993), 121–125.
- [11] D.H. Owens, N. Amann, E. Rogers and M. French, Analysis of linear iterative learning control schemes – A 2D systems/repetitive processes approach, *Multidim. Syst. Signal Process.*, **11**(2000), 125–177.
- [12] J. Wallen, S. Gunnarsson and M. Norrlof, Analysis of boundary effects in iterative learning control, *Int. J. Control*, **86**(2013), 410–415.
- [13] L.V. Hien and H. Trinh, Switching design for suboptimal guaranteed cost control of 2-D nonlinear switched systems in the Roesser model, *Nonlinear Anal. Hybrid Syst.*, **24**(2017), 45–57.
- [14] Z. Xu, H. Su, R. Lu and Z.G. Wu, Non-fragile dissipative filtering for 2-D switched systems, *J. Frank. Inst.*, **354**(2017), 6234–6246.

- [15] H. Trinh and L.V. Hien, On reachable set estimation of two-dimensional systems described by the Roesser model with time-varying delays, *Int. J. Robust Nonlinear Control*, **28**(2018), 227–246.
- [16] J. Liang, F. Wang, Z. Wang and X. Liu, Robust Kalman filtering for two-dimensional systems with multiplicative noises and measurement degradations: The finite-horizon case, *Automatica*, **96**(2018), 166–177.
- [17] L.V. Hien and H. Trinh, Stability analysis and control of two-dimensional fuzzy systems with directional time-varying delays, *IEEE Trans. Fuzzy Syst.*, **26**(2018), 1550–1564.
- [18] L.V. Hien and H. Trinh, Observers design for 2-D positive time-delay Roesser systems, *IEEE Trans. Circuit. Syst.-II: Expr. Brief.*, **65**(2018), 476–480.
- [19] D. Hua, W. Wang, J. Yao and Y. Ren, Non-weighted H_∞ performance for 2-D FMLSS switched system with maximum and minimum dwell time, *J. Frankl. Inst.*, **356**(2019), 5729–5753.
- [20] I. Ghous and Z. Xiang, Robust state feedback H_∞ control for uncertain 2-D continuous state delayed systems in the Roesser model, *Multidim. Syst. Signal Process.*, **27**(2016), 297–319.
- [21] I. Ghous and Z. Xiang, H_∞ control of a class of 2-D continuous switched delayed systems via state-dependent switching, *Int. J. Syst. Sci.*, **47**(2016), 300–313.
- [22] I. Ghous and Z. Xiang, Reliable H_∞ control of 2-D continuous nonlinear systems with time varying delays, *J. Frankl. Inst.*, **352**(2015) 5758–5778.
- [23] L.V. Hien and H. Trinh, A novel approach to exponential stability of continuous-time Roesser systems with directional time-varying delays, *J. Frankl. Inst.*, **354**(2017), 1023–1041.
- [24] H. Trinh and L.V. Hien, Prediction-based approach to stabilization of 2-D continuous-time Roesser systems with directional input delays, *J. Frankl. Inst.*, **357**(2017), 4779–4794.
- [25] A. Kilbas, M. Srivastava and J. Trujillo, *Theory and Applications of Fractional Differential Equations*, Amsterdam, Elsevier, 2006.
- [26] T. Kaczorek and K. Rogowski, *Fractional Linear Systems and Electrical Circuits*, Switzerland, Springer, 2015.
- [27] S. Domek and P. Dworak, *Theoretical Developments and Applications of Non-Integer Order Systems*, Cham, Springer, 2016.
- [28] Y. Ma, J. Lu and W. Chen, Robust stability and stabilization of fractional order linear systems with positive real uncertainty, *ISA Trans.*, **53**(2014), 199–209.

- [29] Z. Wang, D. Yang, T. Ma and N. Sun, Stability analysis for nonlinear fractional-order systems based on comparison principle, *Nonlinear Dyn.*, **75**(2014), 387–402.
- [30] D. Wang and A. Xiao, Dissipativity and contractivity for fractional-order systems, *Nonlinear Dyn.*, **80**(2015), 287–294.
- [31] T. Kaczorek, Fractional 2D linear systems, *J. Autom. Mob. Robotics Intell. Syst.*, **2**(2008), 5–9.
- [32] K. Rogowski, General response formula for fractional 2D continuous-time linear systems described by the Roesser model, *Acta Mech. et Autom.*, **5**(2011), 112–116.
- [33] K. Rogowski, *Positivity Analysis of Continuous 2D Fornasini-Marchesini Fractional Model*, POSTA 16, Switzerland, Springer, 2016.
- [34] L.V. Hien and C.T. Kinh, Decentralised stabilization of positive fractional-order interconnected systems, *IET Control Theory Appl.*, **11**(2017), 2391–2395.
- [35] L.V. Hien and H. Trinh, Exponential stability of two-dimensional homogeneous monotone systems with bounded directional delays, *IEEE Trans. Autom. Control*, **63**(2018), 2694–2700.
- [36] A. Alfidi, A. Hmamed, F. Tadeo and A. Benzaouia, Control with positivity constraint for 2D continuous-time systems in Roesser model, *J. Control Autom. Elect. Syst.*, **32**(2021), 70–81.